C2.2 Neural signalling

NAME	DATE	TOTAL
		_____ / 49

About 50 minutes, or one section at a time. A calculator is allowed. The number of marks shows how many separate points to make. Look at the bold command term: "state" needs a brief answer, while "explain" needs reasons.

Section A: Recall

9 MARKS

1 **DISTINGUISH** between a dendrite and an axon. [C2.2.1] [2]

2 **STATE** the typical resting potential of a neuron. [C2.2.2] [1]

3 **STATE** the number of sodium ions and of potassium ions moved by one cycle of the pump, and the direction of each. [C2.2.2] [2]

4 **IDENTIFY** one neurotransmitter, and one type of synapse where it is used. [C2.2.7] [2]

5 **STATE** the type of cell that forms the myelin sheath in peripheral nerves. [C2.2.4] [1]

6 **STATE** the ion that enters the synaptic knob when the presynaptic membrane depolarizes. [C2.2.6] [1]

Section B: Understanding

16 MARKS

7 **OUTLINE** why a signal can pass in only one direction across a typical synapse. [C2.2.5] [2]

8 **EXPLAIN** how a resting potential is generated and maintained. [C2.2.2] [4]

9 **EXPLAIN** why a nerve impulse is described as electrical. [C2.2.3] [2]

10 **DESCRIBE** how a neurotransmitter is released from a presynaptic membrane. [C2.2.6] [3]

11 **EXPLAIN** how the binding of acetylcholine generates an excitatory postsynaptic potential. [C2.2.7] [4]

12 **STATE** why a neuron uses ATP even when it is not transmitting an impulse. [C2.2.2] [1]

Section C: Working with data

16 MARKS

Values are rounded or invented for practice, based on realistic patterns.

13 [C2.2.4]

The table shows typical conduction speeds in five types of nerve fibre.

FIBRE	DIAMETER / MM	MYELINATED	SPEED / M S^{-1}
SQUID GIANT AXON	500	no	25
COCKROACH GIANT AXON	50	no	10
HUMAN SMALL FIBRE	1	no	1
CAT MOTOR FIBRE	5	yes	30
HUMAN MOTOR FIBRE	20	yes	120

a **CALCULATE** how many times faster the squid giant axon conducts than the human small fibre. [1]

b **DESCRIBE** the effect of diameter on speed in non-myelinated fibres. [1]

c **COMPARE** the cat motor fibre with the cockroach giant axon. [2]

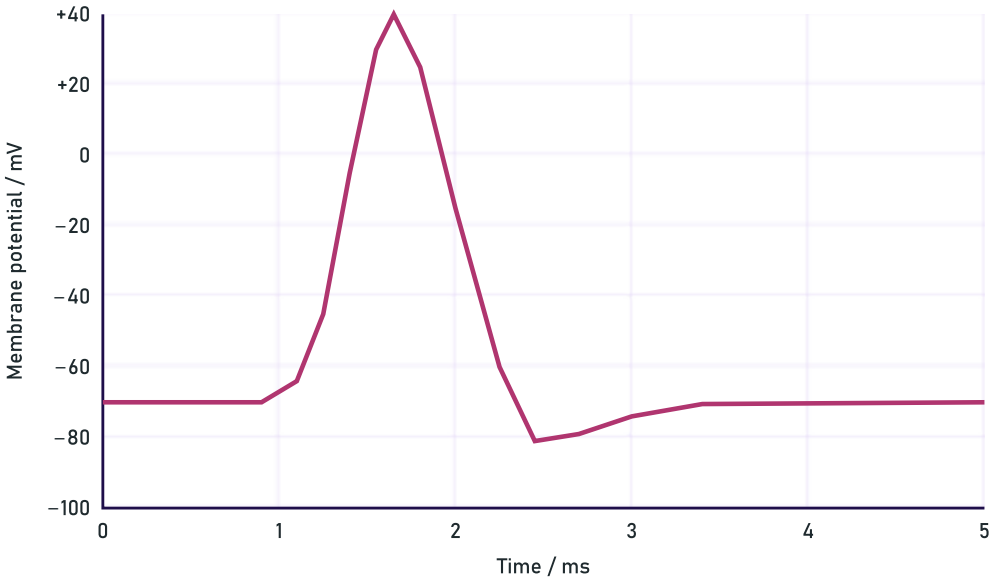
d **EXPLAIN** why a wider fibre conducts faster. [2]

e **EXPLAIN** why myelin increases conduction speed. [2]

f **SUGGEST** an advantage to a squid of having giant axons controlling its escape response. [2]

14 [C2.2.2, C2.2.3]

The graph shows the membrane potential at one point on an axon as an impulse passes.



a STATE the resting potential shown. [1]

b CALCULATE the change in membrane potential from rest to the peak. [1]

c IDENTIFY the ion whose movement causes the rising part of the curve, and its direction. [1]

d IDENTIFY the ion whose movement causes the falling part of the curve, and its direction. [1]

e A second electrode, 60 mm further along the axon, recorded the same peak 3 ms later. CALCULATE the conduction speed in m s^{-1}. [2]

Section D: Longer answers

8 MARKS

15 **DESCRIBE** how a signal passes across a cholinergic synapse, from the arrival of an action potential to the removal of the neurotransmitter. [C2.2.5-7] [8]